

ASSIGNMENT #03

Course: CHE 433: PROCESS SIMULATION WITH ASPEN PLUS
Term: Fall 2025
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Due Date: Friday November 14th, my office (6:00PM)

INSTRUCTIONS FOR ASSIGNMENT DELIVERY

- Prepare a **typed** report of your assignment.
- Include this cover page with your name(s) in the report.
- Upload your report as a PDF file by the Due Date and Time on the UIC Blackboard.
- Upload your Aspen Plus Simulation file (*.apw)
- Bring a hard copy of your report as a PDF file by the Due Date and Time in class

- Assignments that are not typed will not be accepted.
- Late Assignments will not be accepted without prior approval.
- You may use the Internet as well as ChatGPT or other public tools as long as you place a written statement about their use in your assignment report

Problem 1: Biomass Pyrolysis Simulation

Biomass conversion processes that are based on pyrolysis offer high yields of liquid transportation fuels such as gasoline and diesel blend stocks, or refinery intermediates. Fast pyrolysis proceeds by heating the biomass to an intermediate temperature where volatilization begins but the gasification of vapor-phase products does not occur. Fast pyrolysis generates a liquid bio-oil that can be transported efficiently and upgraded to fuels.

In a conceptual process design (see Process Flowsheet Diagram Figure 1. below), the fast pyrolysis reactor system is usually represented as a dual circulating fluidized bed. Biomass and hot sand are introduced into the bottom of a vertical riser reactor. As the solids are carried up the riser by a fluidizing gas, the rapid heating of biomass particles to around 450-500 degrees Celsius, produces non-condensable gases and vapors of oxygenated hydrocarbons. Unreacted biomass flows out as solid char. Solid and vapor phases exit the top of the riser and are separated in a cyclone. Sand and char drop into a char combustor, which is basically a second fluidized bed where the unreacted char is combusted in air to reheat the sand. The energy to drive the endothermic pyrolysis reactions is thus provided via the hot sand, by partial combustion of the biomass itself. The vapor phase is condensed downstream to form the bio-oil. In some cases, the bio-oil separates into organic and aqueous phases. Fast pyrolysis bio-oils can be as high as 50% oxygen by mass and are generally assumed to be hydro-treated downstream, to reduce oxygen.

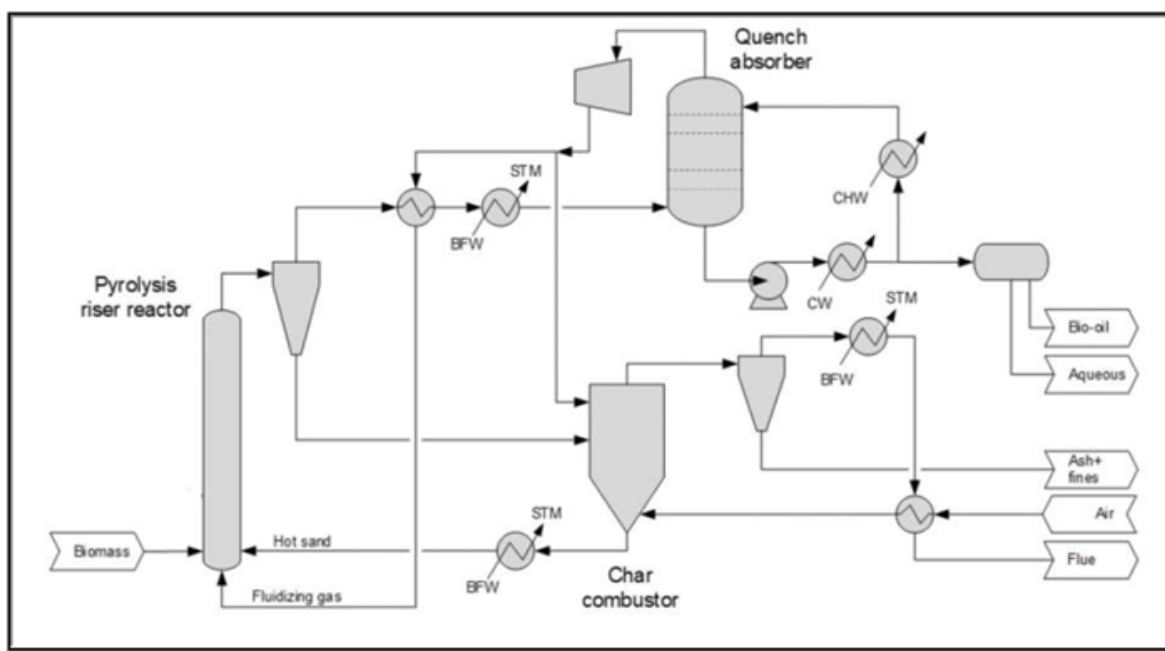


Figure 1: Process Flow Diagram(PFD) of Biomass Fast Pyrolysis

The objective of the assignment is to prepare an ASPEN Plus simulation and calculate:

1. Total Power(KW) consumed
2. Total High(250 °C) and Low(125 °C) pressure steam generated (MMSCFD)
3. Mass yield (%) of incoming carbon to **Bio-oil**
4. Mass yield (%) of incoming carbon to **Flue Gas**
5. Mass yield (%) of incoming carbon to **Waste Water**

Your final report should also include the following

1. Material and Energy Balance with the UIC template
2. A brief outline of your methodology for modeling the pyrolysis reactor
3. A brief outline of your methodology for modeling the char combustor

The Quench Absorber can be simulated using a SEP2 Aspen Plus Unit Operation block

Data for the simulation

1. Bone dry biomass flow rate: **50,000 kg/hr**
2. Biomass feed temperature: **25 deg C**
3. Biomass feed pressure: **2.3 atm**
4. Moisture weight fraction: **0.1**
5. Fluidization gas to Sand mass ratio: **0.1**
6. Bleed Gas to Combustor: **13,000 kg/hr**
7. Bleed Gas temperature: **80 deg C**
8. Bleed Gas pressure: **2.31 atm**
9. Sand to dry biomass mass ratio: **7.0**
10. Biomass **PROXANAL**
 - a. Moisture:
 - b. Fixed Carbon (FC): $0.85 \cdot (100 - \text{ASH})$
 - c. Volatile Matter (VM): $0.15 \cdot (100 - \text{ASH})$
 - d. Ash (ASH): 0.3000
11. Biomass **ULTANAL**
 - e. ASH: 0.30
 - f. CARBON: 49.66
 - g. HYDROGEN: 6.31
 - h. NITROGEN: 0.10
 - i. CHLORINE: 0.00
 - j. SULFUR: 0.08
 - k. OXYGEN: 43.55
12. Biomass **SULFANAL**
 - l. PYRITIC: 0.00
 - m. SULFATE: 0.04
 - n. ORGANIC: 0.04

13. Pyrolysis Reactor Pressure Drop:	0.3 atm
14. Pyrolysis Reactor Temperature:	480 deg C
15. Pyrolysis Reactor Yield (from Experiments)	
o. Gas:	12.9%
i. H2:	0.10%
ii. CO:	5.82%
iii. CO2:	6.03%
iv. CH4:	0.53%
v. C2H4:	0.42%
p. Water:	14.8%
q. Char:	9.7%
r. Organics:	62.6%
i. Acids[Acetic Acid]:	11.1%
ii. Carbonyls[Acetone]:	12.7%
iii. Phenols[m-Cresol]:	4.0%
iv. Methoxyphenols[Guaiacol]:	27.5%
v. Levoglucosan:	4.4%
vi. Furfural:	2.9%

16. Combustor Pressure: **1.12 atm**

17. Combustion products: **NO2, SO2, CO2, H2O**

18. Oxygen in the combustor effluent: **2% mol**

19. Fluidization Gas temperature: **375 deg C**

20. Fluidization Gas pressure: **2.31 atm**

21. Fluidization Gas and Bleed Gas composition, **mol%**

s. CO:	31.19
t. CO2:	20.11
u. Water:	15.09
v. NO2:	0.30
w. SO2:	0.91
x. Methane:	2.62
y. Ethylene:	1.51
z. Acetone:	28.27

22. Sand component: SiO2

23. Temperature of the Quench Absorber (SEP2): 45 deg C

24. Pressure of the Quench Absorber: 1.7 atm

Section #1: PFDs

The full Process flow diagram, which used several hierarchies to hide fake equipment, as well as each hierarchy are shown below.

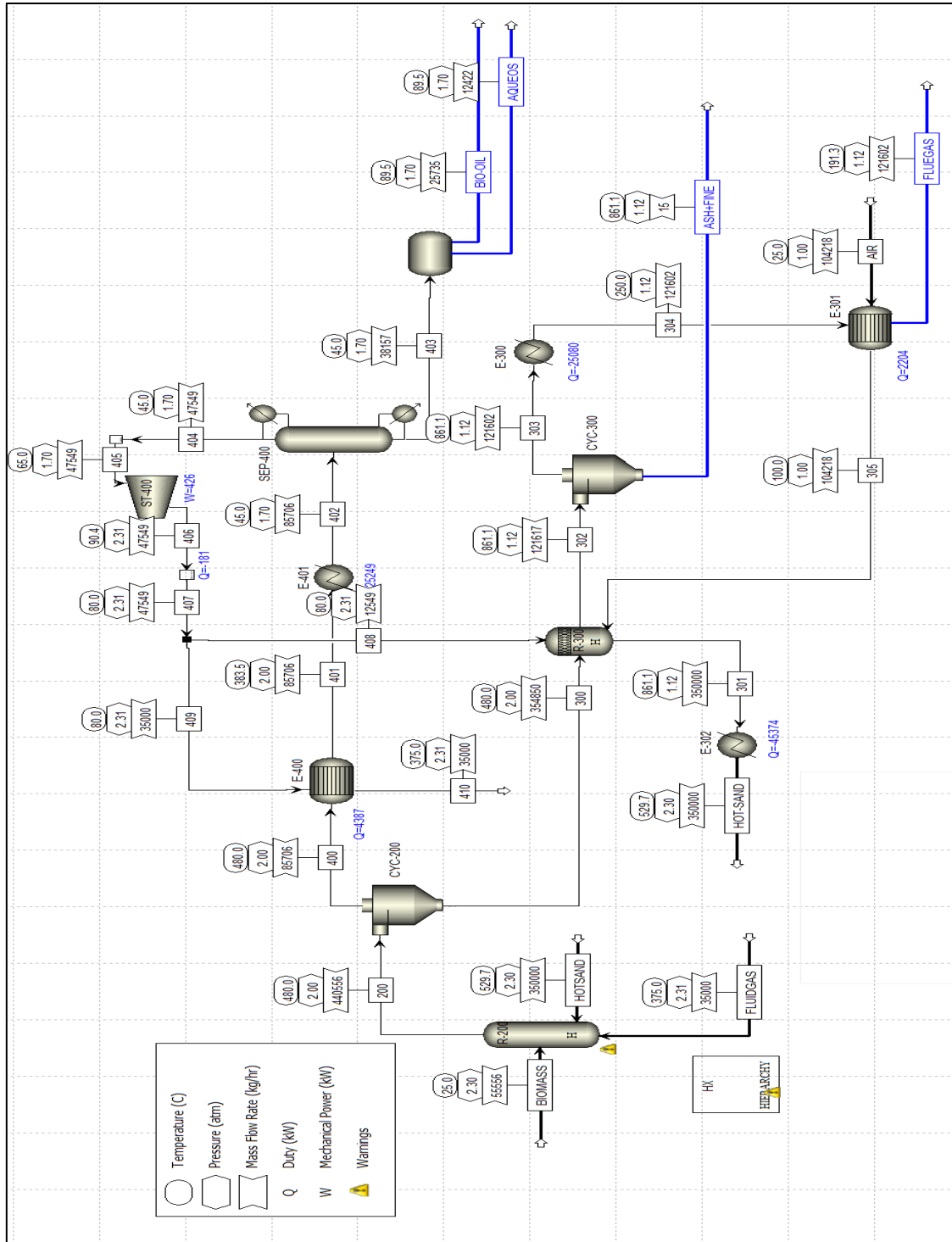
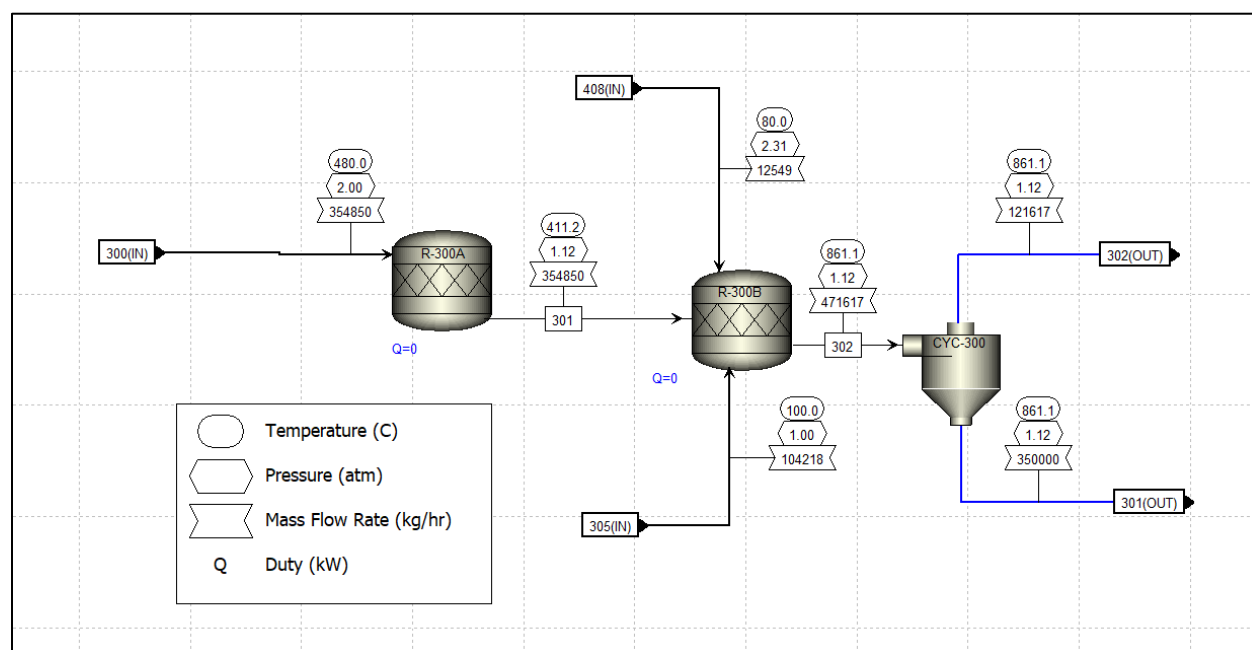
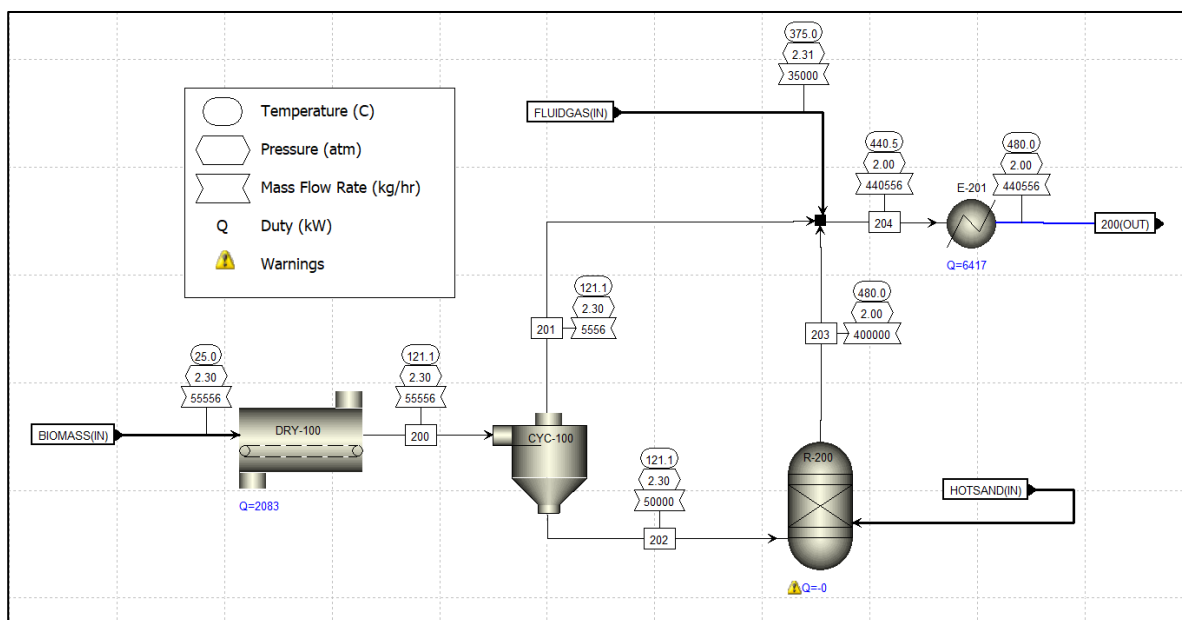
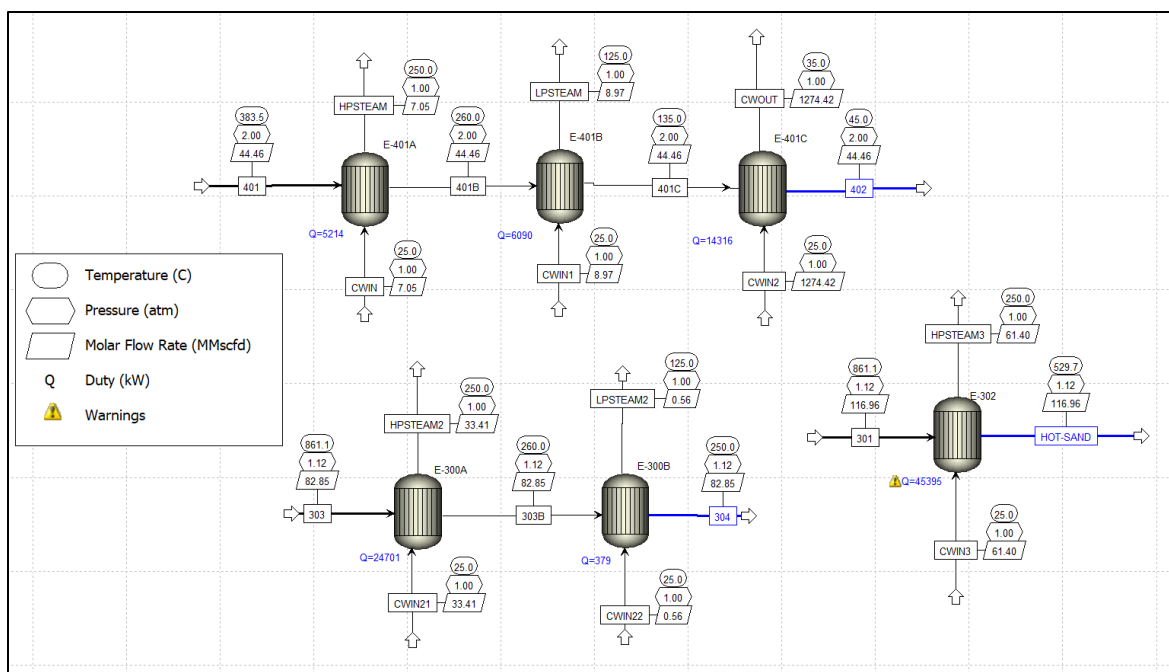


Figure 1: Overall Process Flow Diagram





Section #2: Heat and Weight Balance

The following tables are the stream summaries output from ASPEN V14, in the UIC format.

UIC

Save Save as New Discard Change

Template Stream Summary Options Flows Composition Property Sets Report

General Options Phases Mole Mass Volume Select Properties Display Options Calculation Options Send to Excel/ASW Send to FlowSheet Copy All

Capital: USD Utilities: USD/Year Energy Savings: MW (%) Exchangers - Unknown: 8 OK: 0 Risk: 0 CO₂e Scope 1 32237 kg/hr Scope 2

Main Flowsheet BIOMASS (MATERIAL) - Results (Default)

Material	Vol.% Curves	Wt.% Curves	Petroleum	Polymers	Solids	Status
Description						
- Total Stream						
Temperature	C					
Pressure	atm					
+ Mass Flows	kg/hr					
+ Mass Fractions						
Enthalpy Flow	kW					
Exergy flow rate						
+ MIXED Substream						
+ CISOLID Substream						
+ NC Substream						
<add properties>						

Material	Vol.% Curves	Wt. % Curves	Petroleum	Polymers	Solids	Status			
		Units	400	401	402	403	AQUEOS	BIO-OIL	
Description									
– Total Stream									
Temperature	C		480	383.474	45	45	89.5196	89.5196	
Pressure	atm		2	2	1.7	1.7	1.7	1.7	
+ Mass Flows	kg/hr		85705.5	85705.5	85705.5	38156.7	12421.7	25735	
+ Mass Fractions									
Enthalpy Flow	kW		-125660	-130047	-155297	-81108	-53845.6	-27262.4	
Exergy flow rate									
+ MIXED Substream									
+ CISOLID Substream									
+ NC Substream									
<add properties>									

Table 2: Separation section stream summary (1)

Material	Vol.% Curves	Wt. % Curves	Petroleum	Polymers	Solids	Status				
		Units	404	405	406	407	408	409	410	
Description										
– Total Stream										
Temperature	C		45	65	90.3534	80	80	80	375	
Pressure	atm		1.7	1.7	2.31	2.31	2.31	2.31	2.31	
+ Mass Flows	kg/hr		47548.8	47548.8	47548.8	47548.8	12548.8	35000	35000	
+ Mass Fractions										
Enthalpy Flow	kW		-74510.6	-72761.8	-72335.5	-72516.4	-19138.1	-53378.3	-48990.9	
Exergy flow rate										
+ MIXED Substream										
+ CISOLID Substream										
+ NC Substream										
<add properties>										

Table 3: Separation section stream summary (2)

Material	Vol.% Curves	Wt. % Curves	Petroleum	Polymers	Solids	Status					
		Units	300	301	302	303	304	305	AIR	FLUEGAS	ASH+FINE
Description											
– Total Stream											
Temperature	C		480	861.126	861.126	861.126	250	100	25	191.324	861.126
Pressure	atm		2	1.12	1.12	1.12	1.12	1	1	1.12	1.12
+ Mass Flows	kg/hr		354850	350000	121617	121602	121602	104218	104218	121602	14.55
+ Mass Fractions											
Enthalpy Flow	kW		-1.43617e+06	-1.37824e+06	-74864.4	-74864.6	-99944.9	2196.19	-8.20171	-102149	0.176349
Exergy flow rate											
+ MIXED Substream											
+ CISOLID Substream											
+ NC Substream											
<add properties>											

Table 3: Combustion section stream summary

Section 3: Solution Logic and description

The fast pyrolysis process was modeled in Aspen Plus using the flowsheet and specifications provided in the assignment. Because the conceptual PFD simplifies several operations, the full simulation required additional supporting blocks, assumptions, and organizational structure to reflect realistic process behavior while remaining readable. To keep the final flowsheet clear and consistent with the assignment diagram, three hierarchies were created: one hierarchy for the pyrolysis reactor section, one for the char combustor section, and one for the steam-generation section used to calculate the production of high-pressure and low-pressure steam (reported in MMSCFD). These hierarchies allow the detailed modeling work to remain hidden inside organized subsections, while the top-level flowsheet visually matches the intended simplified PFD.

Properties

The PR-BM property method was selected for this simulation. Several adjustments were made in the Components and NC-Props sections to properly model biomass and solid behavior in Aspen. Two additional components were added: Carbon (as a CI solid) and Sulfur (as a conventional component), which are required for the first RStoic reactor where char is decomposed into elemental species (H_2 , O_2 , Sulfur, Ash, N_2 , and solid Carbon). In addition, ASH was defined as a nonconventional (NC) solid, consistent with typical biomass ash characterization.

All NC solids (WOOD, ASH, and CHAR) were assigned HCOALGEN for enthalpy and DCOALIGT for density. Default option codes were used for all NC components except WOOD, for which the first HCOALGEN option code was set to 6 to enable specification of its heat of combustion. Under Parameters → Pure Components, the heat of combustion (HCOMB) for WOOD was set to 20 MJ/kg, matching common biomass heating values and allowing an adiabatic energy balance to be performed later for the hot sand returning from the combustor.

Although the assignment's PFD illustrates closed loops for sand circulation and fluidizing gas, true recycle loops involving solids can produce strong convergence issues in Aspen. To avoid this, the loops were left open and additional auxiliary equipment was added or specifications were made downstream to emulate the intended loop conditions without compromising solution stability.

methodology for modeling the pyrolysis reactor

The pyrolysis reactor section begins by determining the required wet-biomass feed rate, since the reactor must receive 50,000 kg/hr of bone-dry biomass while the assignment

specifies 10 wt% moisture in the incoming feed. This requires back-calculating the wet-biomass flowrate using

$$\dot{m}_{\text{wet}} = \frac{\dot{m}_{\text{dry}}}{1 - 0.10} \approx 55,556 \text{ kg/hr.}$$

In addition to the flow rate and thermodynamic specifications, the PROXANAL, ULTANAL, and SULFANAL analyses for the biomass were entered exactly as provided in the assignment. The moisture value in PROXANAL was set to 10 wt% to represent the wet-biomass feed. All other values were entered directly as provided. For the char, which appears as a nonconventional component in the reactor effluent, Aspen requires corresponding NC-property data; since no information was provided in the assignment, the char was assumed to have the same PROXANAL, ULTANAL, and SULFANAL composition as dry biomass, but with 0% moisture. This assumption is necessary for Aspen to perform energy and property calculations and is reasonable given the limited information supplied.

The inlet conditions for the biomass (25 °C, 2.3 atm) are taken directly from the problem statement. A DRYER block is used to remove the moisture, with a dryer temperature of 250 °F and a “drying on a wet basis” value of 1×10^{-10} , which ensures essentially complete moisture removal without causing Aspen’s numerical issues associated with setting this value to zero. The dryer output contains a dry NC-solid stream (the bone-dry biomass) and a mixed-phase water stream. These are separated in an SSPLIT block, which separates based on sub streams, and sends the dry biomass (NC) to the reactor and the removed moisture (MIXED) to the downstream mixer.

The hot-sand feed is calculated according to the assignment’s sand-to-dry-biomass mass ratio of 7.0, giving

$$\dot{m}_{\text{sand}} = 7 \times 50,000 = 350,000 \text{ kg/hr.}$$

Likewise, the fluidizing-gas flow is determined from the fluidizing-gas-to-sand ratio of 0.1, giving

$$\dot{m}_{\text{fluidizing gas}} = 0.1 \times 350,000 = 35,000 \text{ kg/hr.}$$

These values are used directly as feed specifications.

The pyrolysis reactor is modeled as an RYIELD block, with mass-fraction yields taken exactly from the assignment. These yields are not atom-balanced, which triggers a warning in Aspen, but this does not affect the rest of the flowsheet since only the product distribution is required. Sand is defined as an inert to prevent unintended participation in the yield model. The reactor operating conditions are fixed at 480 °C with a 0.3 atm pressure drop (entered as -0.3 atm in Aspen).

A mixer is placed immediately after the reactor to combine (1) the reactor effluent, (2) the separated moisture stream, and (3) the fluidizing-gas stream. This mixer is intentionally a “fake” piece of equipment required only to prevent Aspen from forcing the fluidizing gas to participate in the RYIELD decomposition. Because the mixer cools the stream (the fluidizing gas enters at 375 °C and the moisture at 250 °F), a heater is added after the mixer to raise the mixed stream back to 480 °C prior to entering the real cyclone that separates solids from gases. As with all heaters and heat exchangers in the process, no pressure drop is assumed unless explicitly required.

After being reheated to 480 °C, the mixed stream enters an SSPLIT cyclone, where all NC and CI solids are separated from the MIXED-phase vapors; the vapor stream is directed upward toward the quench-absorber section, while the separated solids flow downward into the char-combustor section for further processing.

In the char-combustor hierarchy, a calculator block transfers the sand outlet temperature from the combustor loop back to the sand-inlet stream feeding the pyrolysis reactor. A design specification then adjusts the heater on the sand stream so that this temperature is varied until the pyrolysis reactor operates adiabatically. These details are discussed later in the design-specification section, but they are mentioned here to show how the sand-temperature loop connects the two hierarchies.

methodology for modeling the Quench Absorber

After leaving the solids–gas separator, the vapor stream enters the quench-absorption section, beginning with a heat exchanger that precools the gas prior to absorption. This exchanger is modeled as a simple HeatX block in which the hot pyrolysis vapors transfer heat to the recycled light-gas stream exiting the top of the quench absorber. A zero pressure drop is assumed across this exchanger. The exchanger is specified by setting the cold-stream outlet temperature to 375 °C, because this recycled gas ultimately represents the fluidizing gas entering the pyrolysis reactor. The cold side is the recycled gas stream since its inlet temperature is significantly lower than that of the hot vapors.

Because the Quench Absorber itself was modeled as a SEP2 unit, Aspen does not allow direct specification of absorber operating temperature. Therefore, a heater placed immediately before the SEP2 is used to impose the absorber conditions. This heater was set to 45 °C and 1.7 atm, representing the provided absorber operating conditions. This is a major but reasonable assumption, as the problem statement gives no heater design details. This assumption significantly influences the number of cooling stages required downstream and ultimately determines the amount of LP steam, HP steam, and cooling-water duty generated in later heat-recovery units. These effects are discussed in more detail in the Design Spec and Calculator Block section.

The SEP2 is configured to split components according to the recoveries described verbally by the professor during lecture. All light non-keys (LNKs)— H_2 , N_2 , CO , O_2 , CH_4 , C_2H_4 , CO_2 , SO_2 , and NO_2 —are assigned 100% recovery in the top outlet and represent the recycled light-gas stream. All heavy non-keys (HNKs)—the organics—are assigned 100% recovery in the bottom outlet, which forms the crude liquid product stream. Two components have partial recoveries prescribed explicitly: acetone, which is the light key with 96% recovery in the vapor phase, and water, the heavy key with 20% recovery in the vapor phase. These values were directly supplied by the professor. All other organics (acetic acid, m-cresol, guaiacol, levoglucosan, and furfural) are recovered completely in the liquid phase.

Because the recoveries are specified explicitly, there is no need for the pump + cooler internal recycle loop shown in the assignment's schematic to enhance absorber performance. Instead, the SEP2 alone sets the vapor–liquid distribution.

The bottoms stream from the SEP2 enters a decanter, which separates the aqueous phase from the organic bio-oil. The decanter is operated with zero duty and no pressure drop. The second liquid phase is identified using water as the key component with an efficiency of 100%. Despite specifying perfect phase-split efficiency, a small amount of trace organics (≈ 0.15 wt%) remains in the aqueous phase, while the remainder reports to the organic bio-oil product.

The top vapor stream from the SEP2 is first heated before compression. Because the light-gas mixture at 45°C and 1.7 atm lies below its dew point, the compressor would generate errors if fed directly. Through trial-and-error dew-point checking, the dew point was estimated at approximately 65°C , which is the outlet temperature imposed in the heater preceding the compressor. This heater is visually minimized and disguised on the main flowsheet since it serves only to avoid compressor convergence errors.

The heated vapor stream then enters an isentropic compressor, which raises the pressure to 2.31 atm to match the fluidizing-gas pressure entering the pyrolysis reactor. The outlet of the compressor is fed into another heater, which cools it to 80°C to meet the bleed gas temperature specification provided in the problem statement. Like before, this heater is minimized on the main flow sheet due to it only serving to meet one specification and its small cooling rate. The outlet of the heater is split using a Flow Splitter (FSPLIT) into two streams:

1. Fluidizing-gas stream toward the pyrolysis reactor (not actually closed into a loop, but treated as such conceptually). The FSPLIT is specified to deliver 35,000 kg/hr, following the fluidization-gas-to-sand ratio given in the assignment.
2. Bleed-gas stream toward the char combustor. Although the assignment nominally specifies 13,000 kg/hr, the model yields approximately 12,500 kg/hr, the highest

value achievable without violating other assignment constraints or underfeeding the pyrolysis fluidizing gas.

It is important to note that both split streams contain some H_2 originating from pyrolysis, meaning their compositions differ slightly from the pre-loop molar compositions listed in the assignment. This discrepancy is unavoidable since Aspen cannot remove hydrogen selectively while maintaining the required mass-flow constraints.

The bleed-gas stream leaving the splitter is then fed directly into the char combustor, which begins the next major section of the flowsheet.

methodology for modeling the Char Combustor

After the solids–gases split leaving the pyrolysis section, the sand + char mixture enters the char-combustor hierarchy. This hierarchy contains two sequential RSTOIC reactors that together model (1) char decomposition into elemental species and (2) full combustion with bleed gas and air.

The first RSTOIC block (R-300A) receives char and sand from the cyclone. Char is entered as the single reactant with a stoichiometric coefficient of -1 , and the products— H_2 , O_2 , N_2 , elemental carbon (C-solid), sulfur, and ash—are each assigned an initial stoichiometric coefficient of 1. These coefficients are placeholders only; the true stoichiometric coefficients are later calculated from (1) the ultimate analysis of char (assumed to be the same as dry biomass) and (2) the molecular weights of the product species. These corrected values are imported automatically using a calculator block described later in the report.

Ash composition is defined in its component attributes as 100% ash on a mass basis (an assumption taken from lecture). The reactor is modeled as adiabatic with an operating pressure of 1.12 atm. Sand is inert in this reactor and simply passes through.

The effluent from this first RSTOIC flows to a second RSTOIC (R-300B) that models full combustion. The reactor receives (1) the char-decomposition products, (2) the bleed-gas stream from the quench-absorber recycle loop, and (3) preheated air. To avoid convergence issues in Aspen, NO is specified as the primary combustion product instead of NO_2 . The operating pressure (1.12 atm) and adiabatic assumption are kept consistent with the first reactor.

A cyclone immediately downstream separates C-solid sand from the ash + gas mixture. This cyclone is placed inside the hierarchy so that the main flowsheet shows only the single ash/fines separator depicted in the assignment PFD. The separated sand is routed to a heater outside the hierarchy, where its temperature is adjusted using a design specification

so that the pyrolysis reactor is adiabatic. The heater pressure is manually set to 2.3 atm to match the sand pyrolysis reactor inlet since no unit exists in Aspen to directly control solid-phase pressure.

The gas-phase outlet of the hierarchy (ash + fines + mixed gas) passes to the main flowsheet cyclone, which separates the ash product stream and a gas stream containing N_2 , CO_2 , O_2 , SO_2 , NO_x , H_2O , and minor H_2/CO from upstream sources. This gas enters a heater set to 250 °C. The heater is included to ensure that the inlet temperature to the air-preheat exchanger remains physically reasonable, as the mixture would otherwise exceed 850–900 °C leaving the combustor.

The cooled gas is fed to a heat exchanger (E-301) where ambient air acts as the cold stream. The exchanger specification is the cold-stream outlet temperature, set to 100 °C, which yields an appropriate air-preheat for combustion. The air stream is supplied at 25 °C and 1 atm, with composition assumed as 79% N_2 / 21% O_2 .

The preheated air is sent to the second RSTOIC combustor, since no combustion can occur without O_2 present. To satisfy the requirement that the combustor effluent contains 2 mol% O_2 , a design specification adjusts the inlet air flowrate until this target oxygen fraction is achieved. Keeping O_2 in slight excess avoids failure of the RSTOIC block when insufficient oxygen is supplied for complete combustion, while avoiding unnecessary excess air.

This concludes the char-combustor section, after which the report proceeds to the detailed explanation of the calculator blocks and design specifications that support stoichiometry correction, adiabatic constraints, and sand-loop temperature matching, as well as reporting and calculating deliverables.

Design Specifications

Several design specifications were required to ensure proper operation of the flowsheet, satisfy assignment constraints, and determine steam generation across all heat-exchange stages. These design specifications directly manipulate inlet stream conditions or heat-exchanger cooling-water flowrates so that key temperature, pressure, and composition targets are met.

1. DS-SAND — Adiabatic Pyrolysis Reactor

The first design specification, DS-SAND, forces the pyrolysis reactor to operate adiabatically.

Because the sand loop is not closed in Aspen, the temperature of regenerated sand must be controlled externally. DS-SAND accomplishes this by:

- **Target:** Set the pyrolysis reactor heat duty (QCALC) to 0 kW.

- **Varied variable:** The outlet temperature of heater E-302, which cools the regenerated sand after the combustor.
- **Mechanism:** A calculator block transfers the E-302 outlet temperature into the *sand-to-reactor* inlet stream. DS-SAND then adjusts this temperature until the reactor becomes perfectly adiabatic.

This ensures the sand supplies exactly the heat required to drive the endothermic pyrolysis reactions.

2. DS-AIR — Ensuring Adequate Oxygen for Complete Combustion

The second design specification, DS-AIR, adjusts the combustion-air flowrate entering the second RSTOIC reactor (char combustor).

- **Target:** Achieve 2 mol% O₂ in the combustor effluent.
- **Varied variable:** The inlet molar flow of the AIR stream.

Combustion cannot occur without O₂, and a small excess is required to ensure complete conversion of char, CO, H₂, and other combustibles. The assignment specifies 2 mol% excess O₂, which is used here as a reasonable and safe margin. DS-AIR ensures the RSTOIC block does not fail due to oxygen starvation while avoiding excessive air consumption.

3. DS-E30X — Heat Exchangers for Steam Generation

The three heaters on the main flowsheet (E-401 upstream of the quench absorber, E-300 downstream of the char combustor, and E-302 on the hot-sand loop) were each placed inside a hierarchy and re-modeled as a series of simple heat exchangers. These internal exchangers do not change the overall flowsheet topology; they are used only to convert the sensible cooling of the hot streams into high-pressure (HP) steam, low-pressure (LP) steam, or warmed cooling water. Within each hierarchy, the minimum temperature approach was set to 8 °C so Aspen would not conflict with the 10 °C temperature differences imposed through the design specs below.

For each internal exchanger, a design specification adjusts the cooling-water inlet flowrate so that the utility outlet temperature matches the desired saturation or outlet temperature. The heat exchanger specification is set as the hot stream outlet temperature. This provides a consistent basis for computing the steam generated over each temperature interval.

E-401 hierarchy (three stages: HP steam, LP steam, CW heating)

The gas leaving the main heater E-401 must ultimately enter the quench absorber at 45 °C and 1.7 atm. Inside the hierarchy, this cooling from roughly 380 °C down to 45 °C is split into three exchangers:

1. Stage 1 (HP steam):

- Hot stream cools from ~380 °C to 260 °C.
- Cold side produces HP steam.
- Design spec: vary cooling-water inlet so HP-steam outlet temperature is 250 °C.
- The resulting HP-steam generation is on the order of slightly under 8 MMscfd.

2. Stage 2 (LP steam):

- Hot stream cools from 260 °C to 135 °C.
- Cold side produces LP steam.
- Design spec: adjust cooling-water inlet so LP-steam outlet temperature is 125 °C.
- LP-steam production is slightly higher than the HP-steam from Stage 1 but still modest, also around 8 MMscfd.

3. Stage 3 (cooling water only):

- Hot stream cools from 135 °C to 45 °C.
- No steam is generated; only cooling water is heated.
- Design spec: set cooling-water outlet temperature to 35 °C by changing inlet flow (inlet fixed at 25 °C, 1 atm).
- Because of the relatively small temperature driving force at low temperature and the large duty required to cool to 45 °C, the cooling-water flowrate is extremely large—on the order of 1000+ MMscfd.

Overall, E-401 generates only small amounts of HP and LP steam compared with the very large cooling-water duty required to bring the vapors all the way down to quench-absorber conditions.

E-300 hierarchy (two stages: HP steam, LP steam)

The flue-gas stream entering E-300 is at approximately 860 °C. On the main flowsheet its outlet is fixed at 250 °C, so only HP and LP steam are generated inside the hierarchy:

1. Stage 1 (HP steam):

- Hot stream cools from ~860 °C to 260 °C.

- Generates HP steam.
- Design spec: vary cooling-water inlet so HP-steam outlet temperature is 250 °C.
- This stage produces a moderate amount of HP steam, roughly 30 MMscfd, reflecting the large high-temperature duty available across this interval.

2. Stage 2 (LP steam):

- Hot stream cools from 260 °C to 250 °C.
- Treated as LP-steam generation.
- Design spec: adjust cooling-water inlet so LP-steam outlet is 125 °C.
- Because of the very narrow temperature window, the resulting LP-steam flow is very small, on the order of 0.5 MMscfd.

Thus E-300 is an intermediate contributor to steam production: it provides a noticeable amount of HP steam, but LP steam from this exchanger is almost negligible.

E-302 hierarchy (single stage: HP steam)

The hot sand leaving the char combustor enters E-302 at roughly 529 °C and is cooled to 250 °C before being “sent back” to the pyrolysis reactor. Since this entire cooling range lies above the HP-steam level, only one exchanger is needed inside the hierarchy:

- **Single stage (HP steam):**
 - Hot sand cools from ~529 °C to 250 °C.
 - Cold side produces HP steam.
 - Design spec: adjust cooling-water inlet so HP-steam outlet temperature is 250 °C.
 - Because the sand stream is large and very hot, this stage generates the largest single contribution of HP steam, about 60 MMscfd.

Aspen flags the E-302 exchanger because it involves solid–liquid heat transfer, but the warning is harmless and does not affect the steam-generation computation.

Comparison of steam-generation trends

Comparing the three sections:

- **E-401** produces small amounts of HP and LP steam (≈8 MMscfd each) but an enormous amount of warm cooling water (1000+ MMscfd), because it cools already-partially-cooled vapors all the way down to 45 °C, where steam generation is no longer thermodynamically favorable and most of the duty is in the low-temperature range. This can be mitigated if another specification is set for this

heater that does not bring its outlet temperature below the LP steam temperature range.

- **E-300** generates a moderate HP-steam flow (~30 MMscfd) and almost negligible LP steam (~0.5 MMscfd). This reflects the high inlet temperature of the flue gases but a relatively modest temperature span available for LP-steam generation.
- **E-302** dominates HP-steam production with roughly 60 MMscfd, since it cools a large mass of hot sand entirely within the HP-steam temperature range, with no low-temperature tail that would require cooling water instead.

These trends are physically reasonable: the largest steam production comes from cooling the hottest and heaviest stream (the circulating sand), intermediate steam production comes from the high-temperature flue gases, and the quench-absorber pre-cooler mostly acts as a cooling-water sink, with only modest steam generation because most of its duty occurs at relatively low temperatures.

Calculator Blocks

Four calculator blocks were used in the simulation, each serving a distinct purpose: temperature transfer for the sand loop, steam-flow aggregation, stoichiometric coefficient generation, and carbon-yield calculations. Their functions and internal logic are described below.

C-TEMP — Temperature Transfer for the Sand Loop

The C-TEMP calculator block transfers the sand temperature leaving the char-combustor heater (E-302) to two locations:

1. **The hot-sand inlet to the pyrolysis reactor**, allowing the design specification *DS-SAND* to vary the sand temperature until the pyrolysis reactor becomes adiabatic ($Q_{\text{CALC}} = 0$).
2. **The heat-exchanger surrogate for E-302 inside the steam-generation hierarchy**, ensuring that the HP-steam calculation updates correctly even when the E-302 temperature changes externally.

This is necessary because the sand loop is not physically closed in the flowsheet; therefore Aspen will not automatically propagate temperature between the units unless it is explicitly transferred by a calculator block.

C-STEAM — Summation of Steam Production

The C-STEAM calculator block simply aggregates all steam produced across the model.

Inputs:

- HP Steam Streams: HPSTEAM, HPSTEAM2, HPSTEAM3
- LP Steam Streams: LPSTEAM, LPSTEAM2

The calculator block adds their flow rates (in MMSCFD) and outputs:

- Total HP steam (MMSCFD)
- Total LP steam (MMSCFD)

This block is purely arithmetic and does not feed back into the flowsheet—its outputs are exported only for reporting in the KPIs custom table.

C-RXN — Stoichiometric Coefficients for Char Conversion

The first RSTOIC in the char-combustor hierarchy (R-300A) converts Char into its elemental products: H₂, O₂, N₂, Carbon (C solid), Sulfur, and Ash.

However, the assignment does not provide stoichiometric coefficients, so they must be computed from the ultimate analysis of Char.

Since Char is treated as bone-dry, moisture does not appear in the formula. The stoichiometric coefficient for each product is calculated as:

$$v_i = \frac{\text{ULTANAL}(i)}{100 \times \text{MW}_i}$$

For example, if the ultimate analysis gives 49.66 wt% carbon:

$$v_c = \frac{49.66}{100 \times 12.011}$$

These calculations are performed for C, H, N, S, O, using the imported molecular weights for each elemental or pseudo-elemental product.

The resulting coefficients are exported directly into R-300A's stoichiometric table, replacing the initial unity guesses.

Carbon-Yield Calculator — Bio-Oil, Waste-Water, and Flue-Gas Carbon Yields

The final calculator block computes:

- Carbon yield to bio-oil (%)
- Carbon yield to aqueous phase (%)
- Carbon yield to flue gas (%)

This is done by comparing the mass flow of carbon entering with biomass to the mass flow of carbon leaving in each product stream.

A. Carbon entering with dry biomass

The carbon entering the pyrolysis reactor is:

$$CFLOW = \dot{m}_{\text{biomass}} \times \frac{ULTANAL(C)}{100}$$

This value is exported for later use.

B. Mass Fraction of Carbon in Each Organic Compound

To determine carbon flow in each product stream, the mass fraction of carbon in each compound must be known.

These were calculated manually in Excel from molecular weights imported from Aspen and entered as constants into the calculator block.

Example Calculation (CO₂):

$$X_{C,CO_2} = \frac{12.011}{12.011 + 2(15.999)} = 0.2729$$

As another example, for LVG (C₆H₁₀O₅):

$$X_{C,LVG} = \frac{6(12.011)}{MW_{LVG}}$$

Mass-fraction constants entered for all carbon-bearing components:

- ACETONE
- AACID
- FURFURAL
- MCRESOL
- GUAICOL
- LVG
- CO₂

C. Bio-Oil Carbon Flow

Bio-oil consists mostly of organics, with very trace amounts of water.

For each component of the six organics:

$$\dot{m}_{C,i} = \dot{m}_i \times X_{C,i}$$

Summing:

$$\dot{m}_{C,\text{bio-oil}} = \sum_i \dot{m}_{C,i}$$

Yield:

$$\text{Bio-Oil Carbon Yield (\%)} = 100 \times \frac{\dot{m}_{C,\text{bio-oil}}}{CFLOW}$$

D. Waste-Water Carbon Flow

Opposite to Bio-Oil, the wastewater contains water plus trace organics. The same calculation as above is performed but with much smaller organic flows.

E. Flue-Gas Carbon Flow

Only CO₂ contains carbon:

$$\dot{m}_{C,\text{flue}} = \dot{m}_{CO_2} \times X_{C,CO_2}$$

Yield:

$$\text{Flue-Gas Carbon Yield (\%)} = 100 \times \frac{\dot{m}_{C,\text{flue}}}{\text{CFLOW}}$$

The calculated yields, as well as compressor power and steam generation rates (HP, LP) are reported in a custom table, which can be seen in the next section.

Section #4: Miscellaneous results

The results of the simulation are summarized in the table below, which reports the compressor power, the HP- and LP-steam generation rates, and the carbon-based yields to bio-oil, waste water, and flue gas. The bio-oil yield is the largest, reflecting that a significant fraction of the condensable carbon species formed in the pyrolysis reactor are recovered in the organic liquid phase of the quench-absorber and decanter system. A substantial portion of carbon appears in the flue gas, which is expected since combustion of char and the bleed-gas stream converts a large share of the remaining carbon to CO₂. Only a very small amount reports to the waste-water phase because the aqueous layer contains only dilute concentrations of partially soluble oxygenates. The small remainder of carbon is attributable to trace amounts in solid streams and minor numerical rounding.

The steam-generation results are strongly influenced by the temperature ranges available in each heater. Most of the cooling duties occur well above the HP-steam saturation temperature, so a large fraction of the available sensible heat naturally converts to HP steam. The sand cooler in particular produces the most HP steam because it cools a large, high-mass-flow solid stream from several hundred degrees above the HP-steam level, allowing a substantial amount of energy recovery. The combustor outlet also contributes a moderate amount of HP steam due to its high inlet temperature relative to the 250 °C cutoff. In contrast, the LP-steam window is narrow, both in terms of temperature range and the specific locations where the gases cross into the LP-steam region. As a result, only a small portion of the total duty falls within the 125 °C zone where LP steam can form. Once the gas temperatures fall below the LP-steam range, the remaining heat is too low-grade to generate steam and must instead be rejected to cooling water. Overall, the high proportion

of HP steam reflects both the large thermal duties in the high-temperature zones and the limited opportunities to generate LP steam as the process streams approach quench-absorber conditions.

KPIs			
	Name	Units	Value
▶	Power	kW	426.331
▶	LP-STEAM	MMSCFD	9.52958
▶	HP-STEAM	MMSCFD	101.859
▶	Bio-Oil yield (%)	UNITLESS	62.2802
▶	Waste Water yield (%)	UNITLESS	0.151448
▶	Flue Gas yield (%)	UNITLESS	32.4684

Figure 5: Custom table

Section #5: References

Most information used to solve this problem was derived from the materials provided in the problem description or from lessons taught by Dr. Michael Caracotsios. No external sources or research articles were consulted. Aspen Plus V14 was used to perform the simulation and costing calculations, while ChatGPT was utilized to assist with navigating Aspen and refining the clarity and formatting of this report.

References

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